Combinational Circuits

Combinational Circuits

A combinational circuit is a circuit built by internally *combining* a number of logic gates

this lets it expose only a small number of external connections, while performing a more complex, useful function internally.

Formally:

> A combinational circuit is a circuit with multiple inputs and multiple outputs, in which the output values are uniquely and immediately determined by the current input values

Given the same set of inputs, a combinational circuit will *always* produce the same outputs.

- There is no "history" or "memory" involved - the output at any instant depends only on the input at that same instant.

Combinational Circuits

Not all circuits behave this way.

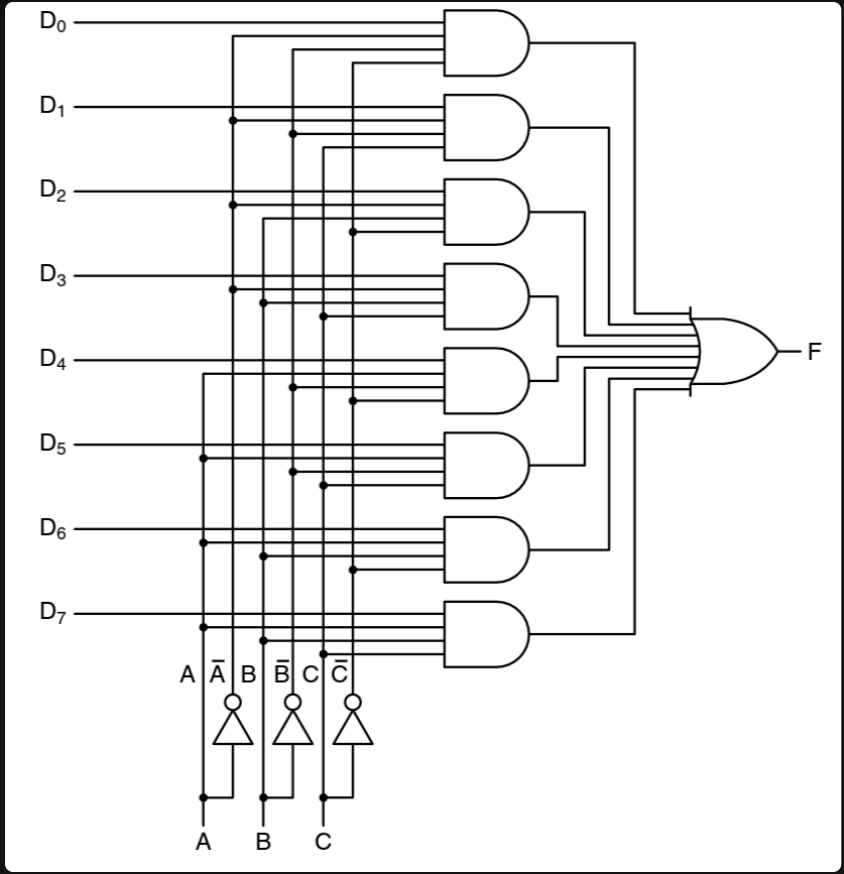
Memory circuits (and sequential circuits more generally) are the counterexample,

Their output depends not just on the current inputs, but also on the circuit's *previous state*.

We'll study these separately from combinational circuits.

	Combinational Circuit	Sequential/Memory Circuit
Output depends on	Current inputs only	Current inputs and past state
Has memory?	No	Yes
Example	Multiplexer, adder	Flip-flop, register

Multiplexers



A multiplexer is a combinational circuit that *selects* one of several input signals and forwards it to a single output.

Formally, a multiplexer is built with:

- 2^n data inputs - the pool of signals.
- n control pins - these encode which one of the 2^n inputs to pass through.
- 1 output - carries the value of whichever input was selected.

Think of it as a controllable switch:

- the n control pins act like an address, and
- whatever binary number they represent determines which of the 2^n inputs gets "*connected*" to the output.

Multiplexers

For example,

- with $n = 2$ control pins,
- there are $2^2 = 4$ data inputs, and
- the 2-bit control value (00, 01, 10, or 11) picks which of the four gets routed to the output.

Exercise

Figure out how this works (in class)

Abstraction

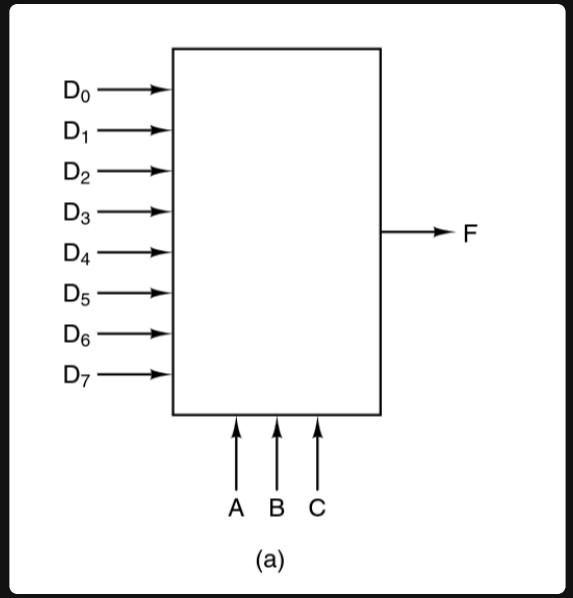
Once we've built and verified a circuit (like a multiplexer) out of individual gates,

we don't need to keep redrawing all of its internal wiring every time we use it.

Instead, we abstract it into a single labeled block:

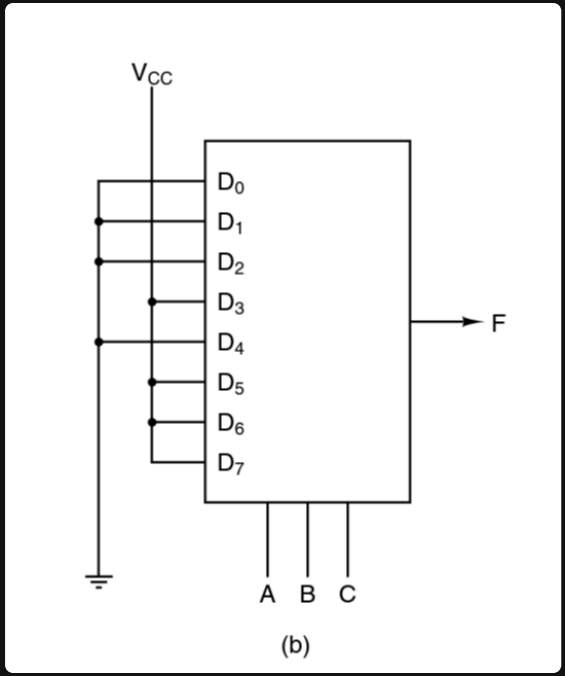
- The internal gate-level detail is *hidden*.
- The block shows only what matters to someone using it: its inputs, its outputs, and its function.

This is usually called *black boxing* and for larger circuits, this is how we'll represent them



The Majority Function, Built from the MUX Abstraction

The *majority function* outputs a 1 if a majority (more than half) of its inputs are 1, and a 0 otherwise.



For 3 inputs, this means: output = 1 whenever at least 2 of the 3 inputs are 1.

We can implement the majority function without designing new gate-level logic:

- Treat the *three* inputs as the *select lines* of a MUX.
- Wire the *data inputs* of the MUX to constants like ground or live (0 or 1), based on what the correct output should be for each combination of the two select inputs.
- The resulting MUX, configured this way, reproduces the truth table of the 3-input majority function exactly.

Demultiplexers (DEMUX)

A demultiplexer performs the *inverse* operation of a multiplexer.

formally, a demultiplexer is built with:

- 1 input - the single signal to be routed.
- n control (select) pins - encode, in binary, which output the input should be sent to.
- 2^n outputs - only one of which receives the input signal at any given time

the rest stay inactive.

> imagine a single signal source and a bank of light bulbs.
> The demultiplexer is like a *switch* that decides *which one* light bulb turns on, based on the control setting
> the "electricity" (the single input) is *routed* to exactly one destination out of many possible ones.

Where a MUX answers "which of many inputs do I let through?", a DEMUX answers "which of many outputs do I send this one input to?"

Exercise

Make a demultiplexer (in class)

Real-World Example: Keyboard Wiring

A standard keyboard has 104 keys.

Wiring each key with its own dedicated pair of connections to the controller chip would be wasteful and impractical. Instead, keyboards use a *matrix layout*:

- Keys are arranged in an 8 × 13 grid:

$$8 \times 13 = 104 \text{ keys}$$

- Instead of 104+ individual wires, the grid only needs:

$$8 + 13 = 21 \text{ wires}$$

The controller scans the grid row by row (or column by column)

effectively using *demultiplexing* logic to select a row/column and multiplexing logic to read back which key(s) are pressed at each intersection.

This is a direct, practical application of the MUX/DEMUX concepts above:

- a small number of control lines (21) can address a much larger number of individual points (104) by exploiting the 2D grid structure,

rather than needing one dedicated wire per key.